

NANYANG TECHNOLOGICAL UNIVERSITY**SEMESTER 2 EXAMINATION 2024-2025****EE7403 – IMAGE ANALYSIS & PATTERN RECOGNITION**

April / May 2025

Time Allowed: 3 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 3 pages.
2. Answer all 4 questions.
3. All questions carry equal marks.
4. This is a closed book examination.
5. Unless specifically stated, all symbols have their usual meanings.

1. An image is modeled by a continuous function $f(x, y) = 2\sin(6\pi x + 8\pi y + 1) + 3$. Its Fourier transform is $F(u, v)$.

- (a) Determine the image
 - (i) maximum grey value M ,
 - (ii) minimum grey value m ,
 - (iii) the positions (u_i, v_i) of all peak values of $|F(u, v)|$ in uv -plane,
 - (iv) the shortest period T of image in xy -plane, and
 - (v) the orientation θ along which the image grey values do not change.

(15 Marks)

- (b) Compute the gradient image in magnitude $|\nabla f(x, y)|$ and direction $\varphi(\nabla f(x, y))$.

(10 Marks)

2. Let $\mathbf{x} = [x_1 \ x_2 \ \dots \ x_n]^T$ be the feature vector to be classified, which has known class-conditional Gaussian PDF for all classes. A minimum weighted distance classifier is expressed by the discriminant function of class ω_i as

$$g_i(\mathbf{x}) = -\sum_{j=1}^n (x_j - \mu_{ij})^2 / \sigma_{ij}^2 + c_i$$

where μ_{ij} and σ_{ij}^2 are the class-conditional mean and variance of x_j for class ω_i . The linear classifier can be formulated by the discriminant function of class ω_i as

$$g_i(\mathbf{x}) = \mathbf{w}_i^T \mathbf{x} + b_i.$$

- (a) Derive the conditions (assumptions) of $\mathbf{x}$, under which the minimum weighted distance classifier can achieve the minimum error rate. (12 Marks)
- (b) Derive the conditions (assumptions) of $\mathbf{x}$ and determine the weighting vector $\mathbf{w}_i$, with which the linear classifier can achieve the minimum error rate. (**Hint:** You may directly use some intermediate results you derived in part (a).) (8 Marks)
- (c) Compare the computational complexity of the above two classifiers for training and for prediction. (5 Marks)

Hint: The general expression of Gaussian PDF is given by:

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{n/2} |\boldsymbol{\Sigma}|^{1/2}} \exp \left[-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right].$$

3. Given a $m \times n$ data matrix $\mathbf{X}$ that contains n centralized training samples, i.e., the sample mean vector is zero.
- (a) Express the covariance matrix $\boldsymbol{\Sigma}$ in terms of the training data $\mathbf{X}$ and write down the definition of the eigenvalues λ_i and eigenvectors ϕ_i of the covariance matrix $\boldsymbol{\Sigma}$. Show that the variance σ_i^2 of the projected training data on the unit-length eigenvector ϕ_i is equal to the eigenvalue λ_i . (8 Marks)
- (b) Prove that two eigenvectors ϕ_k and ϕ_j corresponding to two different eigenvalues λ_k and λ_j , where $\lambda_k \neq \lambda_j$, are orthogonal. (8 Marks)
- (c) Derive the covariance σ_{kj}^2 between the projected training data on the eigenvector ϕ_k and the projected training data on the eigenvector ϕ_j , where $\lambda_k \neq \lambda_j$. (9 Marks)

4. The input of a layer of deep learning network is denoted by $\mathbf{X} = [x_{ij}] = [\mathbf{a}_i] = [\mathbf{b}_j]$, where x_{ij} is a scalar, $\mathbf{a}_i$ is a row vector, $\mathbf{b}_j$ is a column vector and $\mathbf{X}$ is a matrix of size $m \times n$. The output of the layer is also a $m \times n$ matrix, denoted by $\mathbf{Y} = [y_{ij}] = [\mathbf{c}_i] = [\mathbf{d}_j]$, where $\mathbf{c}_i$ is a row vector and $\mathbf{d}_j$ is a column vector. The learnable parameter of this layer is denoted by a matrix of a proper size $\mathbf{W} = [w_{ij}] = [\mathbf{e}_i] = [\mathbf{f}_j]$, where $\mathbf{e}_i$ is a row vector and $\mathbf{f}_j$ is a column vector.
- (a) Suppose this layer of the network generates the output by linearly projecting the input row vector $\mathbf{a}_i$ into another space. Express the output in terms of the input via network parameter in scalar form y_{ij} , vector form $\mathbf{c}_i$ and matrix form $\mathbf{Y}$. What is the size of the learnable parameter matrix $\mathbf{W}$?
(10 Marks)
- (b) Suppose this layer of the network generates the output by linearly projecting the input column vector $\mathbf{b}_j$ into another space. Express the output in terms of the input via network parameter in scalar form y_{ij} , vector form $\mathbf{d}_j$ and matrix form $\mathbf{Y}$. What is the size of the learnable parameter matrix $\mathbf{W}$?
(10 Marks)
- (c) What can you conclude from your results in parts (a) and (b) about the learnable parameters?
(5 Marks)

END OF PAPER

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Please read the following instructions carefully:

1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**
2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.
3. Please write your Matriculation Number on the front of the answer book.
4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.